

Evidence-Guided Visual Retrieval-Augmented Generation with an OCR-free Document Reader



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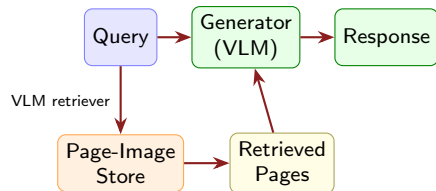
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Introduction

- Much real-world knowledge lives in **visual documents**: PDFs, slides, charts, receipts [4]
- **TextRAG** parses them with OCR and loses layout, tables and figures
- **VisRAG** retrieves and reads pages directly as **images** with a VLM [2]
- Reported **20–40% end-to-end gain** over TextRAG [2]

References: [4, 2]



Visual RAG Architecture

Problem Statement

Text Parsing Loses Visual Information

OCR-based RAG flattens documents into text, dropping layout, charts and tables, so answers that live in the visual structure cannot be recovered.

VLMs Hallucinate Across Multiple Retrieved Images

With top- k pages in context, the model mixes up or invents facts across images instead of grounding each claim in the page it came from [1].

Reading Every Page With a Large VLM Is Expensive

The generator must encode every retrieved image at high resolution; EVisRAG was trained on $8\times$ A100-80GB GPUs, far beyond a regular laptop [1].

References: [1]

Literature Review

Method	Contribution	Limitation
TextRAG [4]	Parses documents to text (OCR), then retrieves and generates	Loses layout, charts and figures
ColPali [5]	Multi-vector retrieval directly over page images	Retrieval only; large multi-vector index
VisRAG [2]	VLM-based retriever and generator over page images	No explicit evidence grounding across images
EVisRAG [1]	Observe → evidence → think → answer, trained with RS-GRPO	Large VLM reads every image; cross-image privacy unstudied
VDocRAG [6]	Unified visual retrieval and generation with document pre-training	Complex visual understanding at high cost
Donut [3]	OCR-free Swin + BART document reader	Single page; no multi-page reasoning or “I don’t know”

Research Gap & Objectives

Research Gap

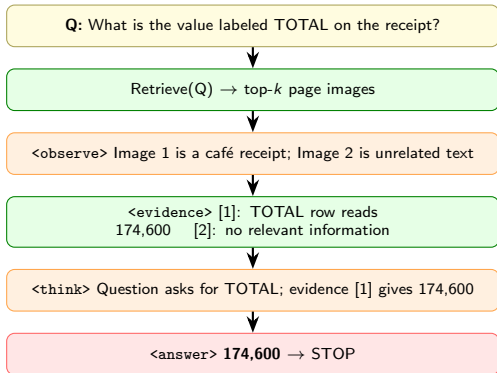
- Large VLM re-reads every retrieved image: slow and memory-heavy [1]
- Linking evidence across images is an unstudied **privacy risk**
- Evaluated only on A100 clusters, not consumer GPUs [1]
- No reliable abstention when evidence is weak

References: [1, 3]

Objectives

- Reproduce light-weight **EVisRAG-3B** on a **Consumer laptop**
- Build a benchmark for **cross-image linkage leakage**
- Propose a **Donut + EVisRAG** hybrid: small reader, large reasoner
- Analyse accuracy, failure modes and cost

EVisRAG: Evidence-Guided Multi-Image Reasoning



Key Steps:

- 1 **Retrieve** top- k page images for the query
- 2 **Observe**: describe what each image shows
- 3 **Evidence**: record per-image facts or “no relevant information”
- 4 **Think**: reason only over recorded evidence
- 5 **Answer**: short final answer

Top-3 images, Qwen2.5-VL-3B/7B backbone [7], SFT (60K) + RS-GRPO (4K) [1, 8]

Base Models: EVisRAG Training & Donut Reader

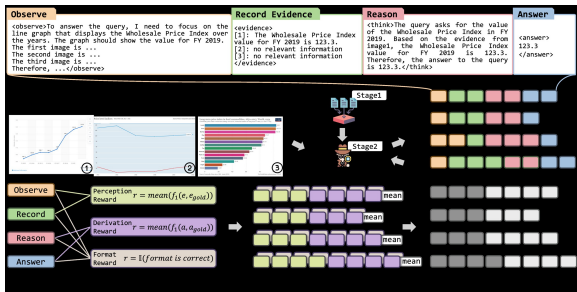


Figure: Sun et al. (2025) [1], github.com/OpenBMB/VisRAG

- **RS-GRPO:** perception, derivation and format rewards are bound to the tokens they score (observe/evidence vs. think/answer) [1, 8]

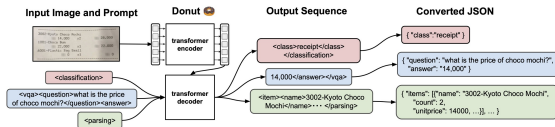
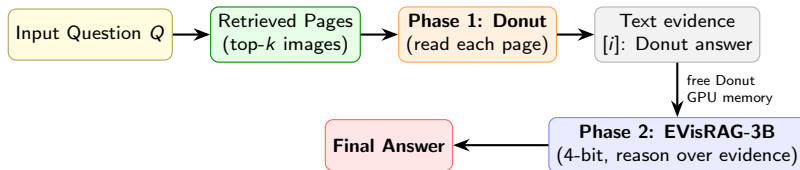


Figure: Kim et al. (2022) [3], github.com/clovaai/donut

- **Donut:** Swin Transformer encoder + BART decoder
- Reads text directly from pixels, no OCR engine
- Pre-trained on SynthDoG + IIT-CDIP (11M images)
- DocVQA: 67.5 at 0.78 s per image [9]

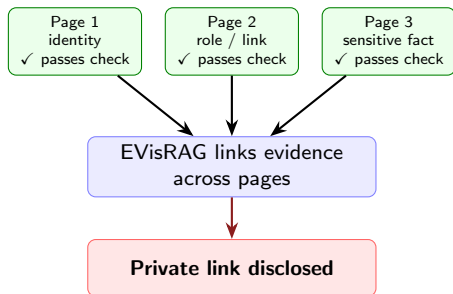
Proposed Methodology: Donut + EVisRAG Pipeline & Setup



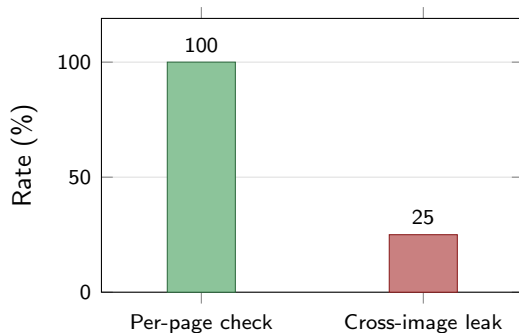
Two sequential phases: the small reader sees every page, the large model only reasons over short text, so both fit on one 6 GB GPU.

Component	Details
Platform	Laptop, 6 GB NVIDIA GPU, Ubuntu on WSL2
Codebase	Official <code>openBMB/VisRAG</code> (EVisRAG) [1] + <code>clovaaai/donut</code> [3]
Reasoner	EVisRAG-3B (Qwen2.5-VL-3B backbone [7]), 4-bit quantized
Reader	Donut, DocVQA fine-tuned (OCR-free) [3, 9]
Data	ChartQA (reproduction), CORD [10] receipt + SynthDoG page (demo), synthetic privacy set
Baseline cost	~1000s per question with full image + text generation

Direction A: Cross-Image Linkage Leakage



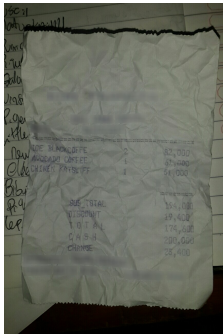
Each page is “safe” alone; the risk appears only when pages are combined



100% of pages pass a per-page privacy check, yet the model leaked the private link in **1 of 4** test cases

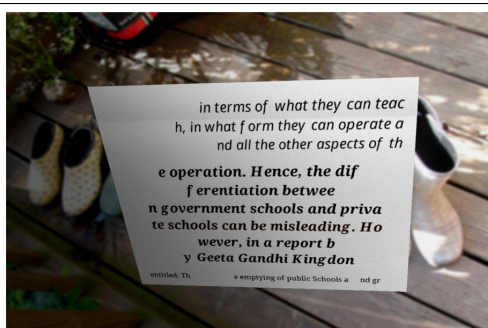
References: [1]

Demo Inputs (Real Document Images)



Demo 1: CORD [10] receipt

Q: What is the value labeled TOTAL? Gold:
174,600



Demo 2: SynthDoG document

Q: Who wrote the report mentioned in the document about schools?

Gold: **Geeta Gandhi Kingdon**

Each question is paired with an irrelevant filler page to test cross-page reasoning

Images: github.com/clovaai/donut (misc/) [3]

Implementation Results (Donut + EVisRAG-3B)

Question (pages)	Donut read (conf.)	EVisRAG-3B	Gold	Correct
TOTAL on the receipt? (2)	174,600 (0.999); filler dropped (0.01)	174,600	174,600	✓
CEO of the company? (3)	maria lindqvist (0.997); receipt dropped (0.70)	maria lindqvist	Maria Lindqvist	✓
Revenue of Asia-Pacific? (3)	536.9	536.9	536.9	✓
TOTAL split between 2 people? (2)	174,600 (0.99)	87,300	87,300	✓
SUBTOTAL – TOTAL? (2, planning)	194,000 and 174,600	19,400	19,400	✓
Capital of Australia? (2)	all pages dropped	insufficient to answer	insufficient	✓
Revenue of region with most employees? (3)	1,845 (wrong row)	1,845	536.9	×
Total (7 questions)				6 / 7

6 of 7 correct 36–90 s per question on an RTX 4050 (6 GB)

References: [1, 3]

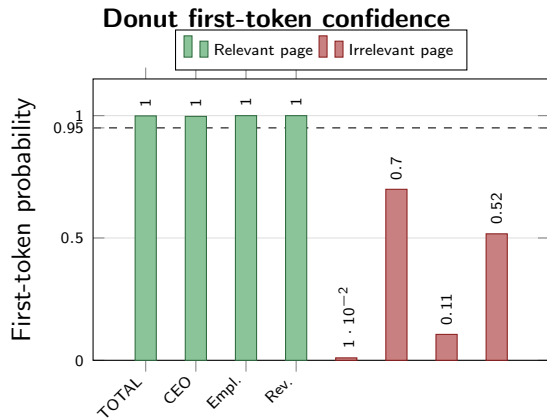
Results: Accuracy Across Datasets (EVisRAG Paper)

Model	ChartQA	InfoVQA	DocVQA	SlideVQA	ViDoSeek	Avg
Qwen2.5-VL-7B [7]	56.32	56.41	53.81	62.23	42.38	54.23
Qwen2.5-VL-32B [7]	70.48	78.41	84.26	79.86	48.16	72.23
EVisRAG-3B [1]	74.96	70.89	78.00	74.46	47.72	69.21
EVisRAG-7B [1]	77.28	79.94	85.45	82.73	52.10	75.50

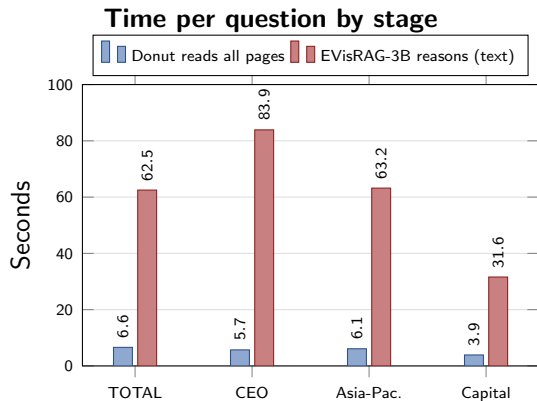
EVisRAG-7B beats its own backbone by 21 points and the $4.5\times$ larger Qwen2.5-VL-32B on every dataset

Source: Sun et al. (2025) [1], Table 1 (accuracy, %); backbone: Bai et al. (2025) [7]

Why It Works: Donut Abstention & Where the Time Goes



Pages below the 0.95 line become “no relevant information”. Donut’s whole-answer probability stays above 0.9 even when it copies unrelated text.



Donut needs about 2–3s per page. Almost all remaining time is 4-bit text generation (about 90–170 tokens); EVisRAG never processes an image.

Conclusion & Future Work

Conclusion

- Donut reads, EVisRAG-3B thinks: runs end to end on a 6 GB laptop GPU (EVisRAG in 4-bit, about 3.9 GB peak)
- 6 of 7 demo questions correct, including arithmetic across Donut's readings (87,300; 19,400) and correct abstention
- Donut's first-token probability gives it the "I don't know" it lacked
- Privacy finding: every page passes a per-page check, yet 1 of 4 cases leaked the linked fact

Future Work

- **Iterative lookups:** feed Donut's first answer into the next lookup to handle row comparisons ("most employees")
- **Speed:** faster 4-bit inference (e.g. llama.cpp) and a measured comparison with image-based EVisRAG on the same GPU
- **Scale evaluation:** accuracy vs. latency on DocVQA / EVisRAG test sets and on real photographed documents
- **Privacy defense:** test the prompt mitigation on a larger benchmark

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Thank You!

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